Chapter 6: Anatomy of Flowering Plants

Dicot Root

Epiblema = outer layer with root hairs **Cortex → Endodermis (Casparian strips) → Stele**
Radial vascular bundle, exarch, 2–4 xylem patches **Pith small & inconspicuous**
Pericycle initiates lateral roots & cambium

Epiblema (Epidermis)

The outermost single layer, made up of barrel-shaped parenchyma cells. Many epiblema cells protrude out to form unicellular **root hairs**. Cuticle is absent.

Cortex

The region between the epidermis and endodermis, made of several layers of thin-walled, round or oval parenchyma cells with intercellular spaces — mainly for storage.

Endodermis

The innermost layer of the cortex; a single layer of barrel-shaped cells without intercellular spaces. Both the tangential and radial walls have a deposition of water-impermeable, waxy suberin in the form of **Casparian strips**.

Stele

All tissue on the inner side of the endodermis — the **pericycle**, **vascular bundles** and **pith** — together constitute the stele.

Pericycle

A single layer of thin-walled parenchymatous cells lying next to the endodermis. Initiation of **lateral roots** and of the **vascular cambium** (during secondary growth) takes place in these cells.

Vascular Bundle

Radial in arrangement. Xylem is **exarch** (protoxylem towards the periphery, metaxylem towards the centre). There are usually **2 to 4 xylem and phloem patches** (monoarch to oligoarch). Later, a cambium ring develops between the xylem and phloem, initiating secondary growth.

Pith

The central axis of the root, made of loosely arranged parenchyma cells with intercellular spaces. In a dicot root the pith region is **small or inconspicuous**.

Conjunctive Tissue & Secondary Growth

The parenchymatous cells lying between the xylem and phloem are called **conjunctive tissue**. In dicot roots a cambium ring later forms here, enabling secondary growth — something monocot roots cannot do.

T.S. of Dicot Root – Regions (outside → inside)

Region	Key Feature
Epiblema	Single layer, unicellular root hairs, no cuticle
Cortex	Several layers, thin-walled parenchyma, intercellular spaces
Endodermis	Single barrel-shaped layer, Casparian strips
Pericycle	Single layer; origin of lateral roots & cambium
Vascular bundle	Radial, exarch, 2–4 xylem patches
Pith	Small, inconspicuous

4 Main Regions of Dicot Root T.S.

- Epiblema (epidermis)
- Cortex
- Endodermis
- Stele (pericycle + vascular bundle + pith)